

Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)

SHAH

SY-146

SY-146

SY-146

OH

Survey: GYRO

Standard Survey Report

12 June, 2019

Survey Report

Company:	Abu Dhabi Company for Onshore Petroleum Operations Ltd(ADCO)	Local Co-ordinate Reference:	Well SY-146
Project:	SHAH	TVD Reference:	RTKB @ 322.0ft
Site:	SY-146	MD Reference:	RTKB @ 322.0ft
Well:	SY-146	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-146	Database:	EDMPRODR5K

Project	SHAH, SHAH		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	Nahrwan - U. Arab Emer.		
Map Zone:	Zone 39N (48 E to 54 E)		

Site		SY-146			
Site Position:		Northing:	2,533,896.00 m	Latitude:	22° 53' 22.801 N
From:	Map	Easting:	790,952.00 m	Longitude:	53° 50' 9.170 E
Position Uncertainty:	0.0 ft	Slot Radius:	13.200 in	Grid Convergence:	1.10 °

Well	SY-146					
Well Position	+N/-S	0.0 ft	Northing:	2,533,896.00 m	Latitude:	22° 53' 22.801 N
	+E/-W	0.0 ft	Easting:	790,952.00 m	Longitude:	53° 50' 9.170 E
Position Uncertainty		0.0 ft	Wellhead Elevation:	ft	Ground Level:	292.1 ft

Wellbore	OH				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	WMM_2005	31/12/2009	1.55	34.96	42,622.27868375

Design				
Audit Notes:				
Version:	Phase:	Tie On Depth:		
Vertical Section:	Depth From (TVD) (ft)	+N/-S (ft)	+E/-W (ft)	Direction (°)

Survey Program	Date	12/06/2019		
From (ft)	To (ft)	Survey (Wellbore)	Tool Name	Description
5,105.0	8,110.0	Survey #5 (OH)	MWD ISCWSA	MWD - Standard

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
224.0	0.25	9.19	224.0	0.5	0.1	0.4	0.11	0.11	0.00
317.0	0.11	19.81	317.0	0.8	0.1	0.7	0.15	-0.15	11.42
411.0	0.03	292.50	411.0	0.9	0.1	0.7	0.12	-0.09	-92.88
504.0	0.05	190.21	504.0	0.8	0.1	0.7	0.07	0.02	-109.99
599.0	0.16	239.38	599.0	0.7	0.0	0.5	0.14	0.12	51.76
692.0	0.04	185.06	692.0	0.6	-0.1	0.4	0.15	-0.13	-58.41
785.0	0.49	148.29	785.0	0.3	0.1	0.2	0.49	0.48	-39.54
873.0	1.46	153.81	873.0	-1.1	0.8	-0.3	1.11	1.10	6.27
946.0	1.82	153.75	946.0	-2.9	1.7	-1.1	0.49	0.49	-0.08

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Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
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Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
1,040.0	1.93	153.72	1,039.9	-5.7	3.1	-2.2	0.12	0.12	-0.03
1,132.0	1.08	146.93	1,131.9	-7.8	4.2	-3.1	0.94	-0.92	-7.38
1,226.0	1.10	148.82	1,225.9	-9.3	5.2	-3.6	0.04	0.02	2.01
1,320.0	0.65	144.06	1,319.8	-10.5	6.0	-4.0	0.48	-0.48	-5.06
1,412.0	0.45	171.86	1,411.8	-11.3	6.3	-4.3	0.36	-0.22	30.22
1,506.0	0.71	145.63	1,505.8	-12.2	6.7	-4.7	0.39	0.28	-27.90
1,599.0	1.02	161.22	1,598.8	-13.4	7.3	-5.2	0.42	0.33	16.76
1,693.0	1.23	141.95	1,692.8	-15.0	8.2	-5.8	0.46	0.22	-20.50
1,786.0	1.41	154.34	1,785.8	-16.8	9.3	-6.5	0.36	0.19	13.32
1,878.0	0.90	160.93	1,877.8	-18.5	10.0	-7.3	0.57	-0.55	7.16
1,971.0	0.55	177.39	1,970.8	-19.7	10.3	-8.0	0.43	-0.38	17.70
2,065.0	0.65	200.65	2,064.7	-20.6	10.1	-8.8	0.28	0.11	24.74
2,160.0	0.45	203.34	2,159.7	-21.5	9.8	-9.6	0.21	-0.21	2.83
2,253.0	0.43	314.61	2,252.7	-21.6	9.4	-10.0	0.78	-0.02	119.65
2,345.0	2.13	19.14	2,344.7	-19.7	9.7	-8.4	2.16	1.85	70.14
2,439.0	6.15	13.46	2,438.5	-13.1	11.4	-2.3	4.29	4.28	-6.04
2,534.0	9.59	9.52	2,532.5	-0.4	13.9	8.9	3.66	3.62	-4.15
2,626.0	11.47	14.95	2,623.0	16.0	17.6	23.6	2.31	2.04	5.90
2,720.0	13.43	22.24	2,714.8	35.1	24.1	42.3	2.67	2.09	7.76
2,814.0	15.67	25.88	2,805.8	56.7	33.8	64.8	2.57	2.38	3.87
2,907.0	18.06	31.53	2,894.8	80.3	46.8	91.1	3.11	2.57	6.08
3,000.0	18.07	25.87	2,983.2	105.5	60.6	119.2	1.89	0.01	-6.09
3,093.0	18.26	22.18	3,071.6	132.0	72.4	146.9	1.25	0.20	-3.97
3,186.0	19.24	18.44	3,159.6	160.0	82.8	174.8	1.67	1.05	-4.02
3,280.0	21.73	16.72	3,247.7	191.4	92.7	204.8	2.72	2.65	-1.83
3,373.0	24.85	16.02	3,333.1	226.7	103.0	238.1	3.37	3.35	-0.75
3,465.0	29.13	14.09	3,415.0	267.0	113.8	275.5	4.75	4.65	-2.10
3,559.0	31.22	15.42	3,496.3	312.7	125.9	317.7	2.33	2.22	1.41
3,652.0	31.98	16.29	3,575.5	359.5	139.2	361.7	0.95	0.82	0.94
3,746.0	38.23	16.82	3,652.4	411.3	154.6	410.7	6.66	6.65	0.56
3,849.0	44.01	15.62	3,729.9	476.4	173.5	472.0	5.66	5.61	-1.17
3,932.0	46.65	17.26	3,788.3	533.0	190.2	525.5	3.48	3.18	1.98
4,026.0	49.86	20.53	3,850.9	599.3	212.9	590.2	4.29	3.41	3.48
4,120.0	46.76	22.77	3,913.4	664.5	238.8	656.3	3.75	-3.30	2.38
4,213.0	47.34	25.54	3,976.8	726.6	266.7	721.3	2.27	0.62	2.98
4,298.0	52.32	27.51	4,031.6	784.7	295.7	784.0	6.12	5.86	2.32
4,328.0	52.85	28.10	4,049.8	805.8	306.8	807.2	2.36	1.77	1.97
4,428.0	52.85	28.10	4,110.2	876.1	344.3	884.7	0.00	0.00	0.00
5,023.7	87.66	40.12	4,309.0	1,328.5	658.6	1,431.9	6.13	5.84	2.02
5,105.0	87.70	42.09	4,312.3	1,389.7	712.1	1,513.2	2.42	0.05	2.42
5,169.0	88.60	43.57	4,314.4	1,436.6	755.6	1,577.1	2.71	1.41	2.31
5,260.0	89.95	40.92	4,315.5	1,504.0	816.7	1,668.1	3.27	1.48	-2.91

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Well:	SY-146	North Reference:	Grid
Wellbore:	OH	Survey Calculation Method:	Minimum Curvature
Design:	SY-146	Database:	EDMPRODR5K

Survey									
Measured Depth (ft)	Inclination (°)	Azimuth (°)	Vertical Depth (ft)	+N/-S (ft)	+E/-W (ft)	Vertical Section (ft)	Dogleg Rate (°/100ft)	Build Rate (°/100ft)	Turn Rate (°/100ft)
5,353.0	87.73	39.56	4,317.4	1,574.9	876.8	1,761.0	2.80	-2.39	-1.46
5,446.0	86.93	38.66	4,321.7	1,647.0	935.4	1,853.9	1.29	-0.86	-0.97
5,540.0	86.74	39.78	4,326.9	1,719.7	994.7	1,947.7	1.21	-0.20	1.19
5,632.0	87.24	41.12	4,331.7	1,789.6	1,054.3	2,039.5	1.55	0.54	1.46
5,728.0	88.29	40.06	4,335.5	1,862.5	1,116.7	2,135.4	1.55	1.09	-1.10
5,819.0	87.42	39.54	4,338.9	1,932.3	1,174.9	2,226.3	1.11	-0.96	-0.57
5,913.0	86.99	45.18	4,343.5	2,001.7	1,238.2	2,320.2	6.01	-0.46	6.00
6,006.0	86.66	42.93	4,348.6	2,068.4	1,302.7	2,412.9	2.44	-0.35	-2.42
6,099.0	89.03	44.89	4,352.1	2,135.4	1,367.2	2,505.7	3.31	2.55	2.11
6,194.0	86.44	43.52	4,355.9	2,203.4	1,433.4	2,600.6	3.08	-2.73	-1.44
6,287.0	88.60	42.67	4,359.9	2,271.2	1,496.8	2,693.4	2.50	2.32	-0.91
6,380.0	88.97	44.14	4,361.9	2,338.8	1,560.7	2,786.3	1.63	0.40	1.58
6,473.0	88.04	43.79	4,364.3	2,405.7	1,625.3	2,879.2	1.07	-1.00	-0.38
6,566.0	89.77	38.00	4,366.1	2,476.0	1,686.1	2,972.1	6.50	1.86	-6.23
6,662.0	91.13	34.01	4,365.3	2,553.6	1,742.5	3,067.7	4.39	1.42	-4.16
6,760.0	90.26	40.64	4,364.1	2,631.5	1,801.9	3,165.4	6.82	-0.89	6.77
6,847.0	89.59	38.13	4,364.3	2,698.7	1,857.1	3,252.3	2.99	-0.77	-2.89
6,939.0	90.39	37.36	4,364.3	2,771.5	1,913.4	3,344.1	1.21	0.87	-0.84
7,033.0	89.21	34.75	4,364.6	2,847.4	1,968.8	3,437.7	3.05	-1.26	-2.78
7,127.0	89.15	32.85	4,365.9	2,925.5	2,021.0	3,530.9	2.02	-0.06	-2.02
7,220.0	89.59	38.28	4,367.0	3,001.2	2,075.1	3,623.3	5.86	0.47	5.84
7,313.0	89.34	40.94	4,367.8	3,072.8	2,134.4	3,716.3	2.87	-0.27	2.86
7,354.0	89.28	40.90	4,368.3	3,103.8	2,161.2	3,757.3	0.18	-0.15	-0.10
7,407.0	90.94	42.19	4,368.2	3,143.4	2,196.4	3,810.3	3.97	3.13	2.43
7,499.0	89.77	35.56	4,367.7	3,215.0	2,254.1	3,902.1	7.32	-1.27	-7.21
7,542.0	90.94	37.51	4,367.4	3,249.6	2,279.7	3,945.0	5.29	2.72	4.53
7,592.0	89.83	43.68	4,367.1	3,287.5	2,312.2	3,994.9	12.54	-2.22	12.34
7,686.0	88.16	48.08	4,368.7	3,352.9	2,379.7	4,088.6	5.01	-1.78	4.68
7,779.0	86.86	43.51	4,372.8	3,417.7	2,446.3	4,181.2	5.10	-1.40	-4.91
7,873.0	90.56	42.89	4,374.9	3,486.2	2,510.6	4,275.1	3.99	3.94	-0.66
7,966.0	90.57	44.40	4,373.9	3,553.5	2,574.8	4,368.1	1.62	0.01	1.62
8,051.0	88.35	39.39	4,374.8	3,616.7	2,631.5	4,453.0	6.45	-2.61	-5.89
8,110.0	90.12	37.23	4,375.5	3,663.0	2,668.1	4,511.9	4.73	3.00	-3.66

Casing Points				
Measured Depth (ft)	Vertical Depth (ft)	Name	Casing Diameter (in)	Hole Diameter (in)
205.0	205.0	CONDUCTOR	13.375	13.375
4,315.0	4,041.9	INTERMEDIATE CASING	9.625	12.250
5,111.9	4,312.6	PRODUCTION LINER	7.000	8.500

Survey Report

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Checked By: _____	Approved By: _____	Date: _____
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